

From Output Quality to Behavioral Validity: A Structured Review and Framework for Repurposed Interactive LLM Systems

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Abstract

Large language models (LLMs) are increasingly used as counseling agents, simulated clients, and patient agents, raising questions about how their behavior is evaluated across interaction. We examine evaluation practices through a structured review of 52 papers on LLM-based counseling, therapy, and behavioral simulation. The review identifies three recurring patterns. First, theory grounding is common, but its translation into evaluation measures does not by itself establish construct validity. Second, extended dialogue is common, but evaluation often focuses on individual responses or aggregate patterns rather than behavioral trajectories. Third, main empirical conclusions are generally supported, but claim-evidence alignment weakens as conclusions extend toward human-facing outcomes. We synthesize these findings into *Behavior-Grounded Evaluation*, an empirically grounded framework that treats evaluation as an evidence chain linking behavioral claims to constructs, observable markers, trajectories, interventions, evaluators, and evidence boundaries. A case study shows how the framework could extend an interactive LLM evaluation to assess the evidential support for its behavioral claims. Our findings suggest stronger evaluation requires not simply more metrics or evaluators, but evidence that the intended behavior is measured validly, develops appropriately across interaction, and supports the scope of the claims being made.

1 Introduction

Large language models (LLMs) are increasingly repurposed as interactive systems, including counseling agents, simulated clients or patients, and role-play partners (Wang et al., 2024b, 2025c; Lee et al., 2024; Sungarda et al., 2025). Unlike systems evaluated primarily on static inputs and outputs, these applications make claims about *behavior*: whether a simulated client maintains a profile, whether disclosure or resistance changes appropriately in re-

sponse to counselor actions, or whether a counseling agent maintains therapeutic stance across interaction. We use *behavioral validity* to refer to whether an interactive system actually exhibits its claimed behavioral patterns across interaction.

This creates an evaluation-alignment problem: evidence about individual outputs may not establish claims about behavior across interaction. Consider a simulated client claimed to remain guarded early in counseling and gradually disclose more as rapport develops. An evaluation that rates individual responses for realism, appropriateness, or role consistency may establish response quality, but it cannot by itself establish the claimed behavioral trajectory. A sequence of plausible responses may still drift from the client profile or respond inconsistently to counseling strategies across turns (Yang et al., 2025b; Wang et al., 2026). The central challenge is therefore to evaluate whether the claimed behavior is actually exhibited, maintained, or changed appropriately across interaction.

LLM-based counseling, therapy, and role-play simulation provide a useful stress case because many of their target constructs—including empathy, alliance, resistance, and disclosure—are inherently interactional (Li et al., 2024; Kim et al., 2025c). Prior reviews and evaluation frameworks for mental-health AI have highlighted broader concerns around reliability, clinical applicability, standardized evaluation, user experience, and longitudinal evidence (Hua et al., 2025; Bucher et al., 2025; Shao et al., 2025; Arnaout et al., 2026). Our focus is the evaluation problem that arises when LLMs are repurposed as interactive systems: *what should an evaluation test when the system is claimed to exhibit behavior that develops across interaction?*

We examine this problem through a structured empirical review of 52 papers on LLM-based counseling, therapy, role-play, and behavioral simulation. The review yields three connected diagnoses: theory grounding is common, but its translation

into evaluation measures does not by itself establish construct validity; extended dialogue is common but trajectory-level evaluation remains limited; and although main empirical conclusions are generally supported, claim–evidence alignment weakens as conclusions extend toward broader behavioral or human-facing outcomes.

We synthesize these findings into *Behavior-Grounded Evaluation*, an empirically grounded framework organized around three questions: *What behavior is being measured? How should that behavior develop across interaction? What claims can the resulting evidence support?* These questions are operationalized through seven linked components: *Claim* → *Construct* → *Behavioral Markers* → *Trajectory* → *Intervention* → *Evaluator* → *Evidence Boundary*. Together, these components form an evidence chain linking behavioral claims to what is measured, how behavior develops across interaction, and what conclusions the resulting evidence can support.

This paper makes three contributions. First, we provide a structured empirical diagnosis of evaluation practice across the 52-paper corpus, showing where evaluation remains limited across construct measurement and behavioral trajectories, and how claim–evidence alignment changes as conclusions broaden. Second, we introduce **Behavior-Grounded Evaluation**, an empirically grounded framework for aligning behavioral claims with construct measurement, interaction trajectories, evaluators, and evidence boundaries. Third, through a case study, we show how the framework can extend an existing interactive LLM evaluation to examine behavioral drift, intervention sensitivity, and the evidential scope of resulting conclusions. Together, these contributions provide an empirically grounded methodology for moving from evaluation of individually plausible outputs toward evaluation of behavioral validity in interactive LLM systems.

2 Related Work: From Mental-Health AI Evaluation to Behavioral Evaluation

Prior reviews consistently show that evaluating mental-health LLMs and conversational agents requires more than model- and response-level metrics. Across mental-health conversational agents, psychotherapy, training, and clinical communication, prior work identifies recurring concerns around clinical relevance, standardized evaluation, user outcomes, longitudinal evidence, and the need

to account for dynamic and contextual interaction (Cho et al., 2023; Hua et al., 2025; Atapattu et al., 2025; Na et al., 2025; Bucher et al., 2025; Wang et al., 2025b; Shao et al., 2025). Together, this literature establishes the need for contextual and clinically meaningful evaluation of interactive mental-health systems.

Recent work has also proposed more explicit evaluation frameworks. Wallach et al. (2025) frame generative-AI evaluation as a measurement problem, linking construct definition, operationalization, measurement instruments, and validity. In mental health, Flathers et al. (2025) propose a clinical framework spanning technical safety, healthcare knowledge, and clinical reasoning, while Arnaout et al. (2026) propose a broader responsible-evaluation framework organized around validity, reliability, implementation, and maintenance. Complementary work evaluates downstream effectiveness, including meta-analytic evidence on human mental-health outcomes from generative-AI chatbot interventions (Zhang et al., 2025b).

Building on these foundations, we address a specific problem that arises when LLMs are repurposed as interactive behavioral systems: *how evaluation should establish the behavior attributed to the system across interaction, and what conclusions the evidence can support*. We therefore examine construct measurement, behavioral trajectories, evaluator grounding, and claim–evidence alignment as connected parts of behavioral evaluation. Table 1 positions this contribution relative to closely related reviews and evaluation frameworks.

3 Method

We constructed and coded a corpus of LLM-based counseling, therapy, role-play, and psychologically grounded behavioral-simulation papers.

3.1 Corpus construction

We searched ACL Anthology, ACM Digital Library, Google Scholar, and targeted HCI journals, including IJHCS and IJHCI, to cover NLP, HCI, and adjacent digital-health venues. Queries combined terms for LLMs and generative agents with counseling, psychotherapy, mental health, patient/client simulation, and evaluation. The search returned 900 records; after duplicate removal and title/abstract screening, 112 underwent full-text assessment, yielding a final corpus of 52 papers. Figure 1 summarizes the process.

Prior work and focus	Relation to our contribution
Mental-health LLM and conversational-agent reviews: response quality, clinical relevance, safety, user outcomes, and longitudinal evidence Cho et al., 2023 ; Hua et al., 2025 ; Atapattu et al., 2025 ; Bucher et al., 2025	Identify broad gaps in clinical relevance, safety, user outcomes, and longitudinal evidence. Our review focuses specifically on how behavioral claims are evaluated, examining construct measurement, behavioral trajectories, evaluator grounding, and the scope of evidence used to support resulting conclusions.
Dynamic, contextual, and outcome-oriented evaluation: psychotherapy process, interaction context, longitudinal assessment, and human outcomes Na et al., 2025 ; Wang et al., 2025b ; Shao et al., 2025 ; Zhang et al., 2025b	Motivate evaluation beyond isolated responses toward interaction dynamics and downstream human outcomes. Our review examines whether interactive behavior is evaluated across trajectories and meaningful interventions, and how claim–evidence alignment changes as conclusions broaden beyond directly evaluated outcomes.
General and responsible evaluation frameworks: construct validity, measurement, safety, reliability, and implementation Wallach et al., 2025 ; Flathers et al., 2025 ; Arnaout et al., 2026	Provide broader foundations for measurement, clinical validity, and responsible evaluation. Our framework focuses specifically on behavioral claims in interactive LLM systems, linking constructs and markers to trajectories, interventions, evaluator grounding, and explicit evidence boundaries.
Our work: Behavior-Grounded Evaluation	Connects what behavior is measured, how it develops across interaction, and what conclusions the resulting evidence can support.

Table 1: Positioning relative to closely related reviews and evaluation frameworks.

3.2 Inclusion criteria

We included papers that (1) studied LLM-based or generative-agent systems; (2) involved counseling, therapy, mental-health support, patient/client simulation, or psychologically grounded behavioral simulation; and (3) reported evaluation of interaction, simulation or counseling behavior, user experience, safety, or model performance. We excluded classification- or detection-only work, systems without interactive generation or simulation, work outside this scope, and papers without accessible full text.

3.3 Coding scheme and validation

We iteratively developed a 40-field codebook covering simulation design, evaluation methods, human-centered targets, theory grounding, reliability/transparency, and claim–evidence alignment. Key dimensions captured interaction dynamics and evaluation scope, evaluator type and validation, theory grounding, and human-facing evidence.

Two researchers independently coded the papers using the shared codebook. Before adjudication, mean raw agreement across fields was 82.5%; for fields where Cohen’s κ was computable, mean κ

was 0.61. Disagreements and ambiguous cases were resolved through discussion with the faculty lead. The complete 52-paper list and 40-field codebook are provided in Appendix A and Appendix B.

4 Field Diagnosis: What Evaluation Measures and Misses

This section reports three recurring patterns across the 52-paper corpus, organized around three questions: (1) does evaluation measure the intended construct (e.g., resistance rather than negative language)? (2) does it capture how the construct develops across interaction (e.g., whether resistance persists across turns rather than appearing in individual responses)? (3) do conclusions remain commensurate with the evidence, particularly when extending beyond what was directly evaluated? Together, these questions concern *construct measurement*, *behavioral trajectory*, and *evidence boundaries*.

4.1 Theory reaches evaluation, but construct validity requires separate scrutiny

Theory is widely incorporated into evaluation. Forty of 52 papers showed strong theory grounding and another 10 partial grounding; 30 strongly

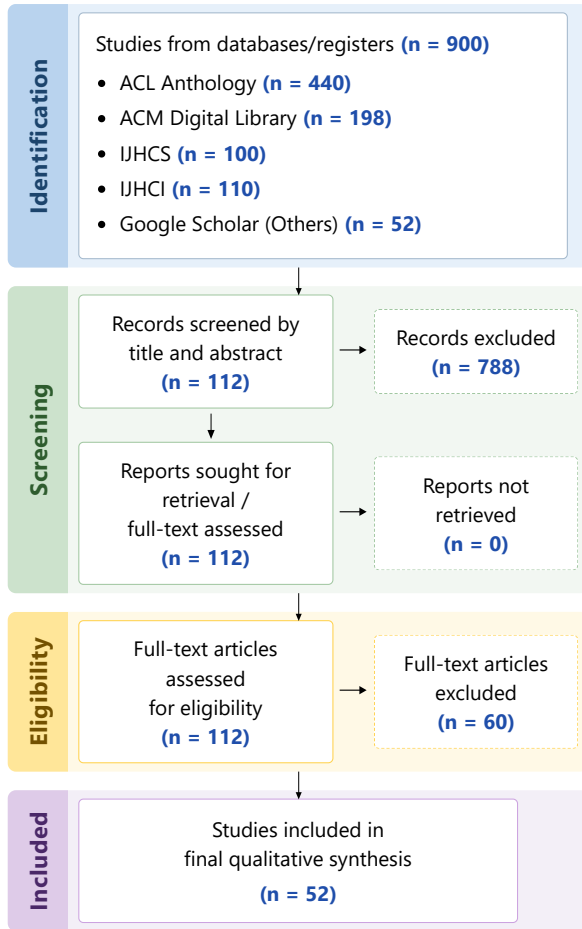


Figure 1: Corpus construction and screening process.

operationalized theory-related constructs, 16 did so partially, and 6 showed no clear theory-shaped evaluation (Appendix Tables 18, 19, and 20).

Theory-related constructs were assessed through psychological scales, therapy coding systems, theory-specific benchmarks, and construct-specific rubrics, including measures of therapeutic alliance and counseling skills (Wang et al., 2026, 2024a), motivational interviewing (Kim et al., 2025a; Yang et al., 2025a), and CBT-oriented behavior (Lee et al., 2024). Across the corpus, standardized psychological or clinical scales appeared in 9 papers and established therapy or counseling coding systems in 7, compared with theory-specific rubrics in 35 and benchmark tasks in 26 (Appendix Table 21). These distributions show that theoretical constructs are translated into evaluation measures through different approaches. However, the use of an established scale, benchmark, or custom rubric does not by itself demonstrate that the resulting scores validly capture the intended construct.

A related issue concerns how these measures are scored. Automatic metrics appeared in 35 pa-

pers, human experts in 34, and LLM judges in 30; 15 papers combined all three (Appendix Tables 14 and 15). Examples include expert or clinician ratings combined with LLM-based rubric scoring (Qiu et al., 2025; Aghakhani et al., 2025; Cai et al., 2026) and task-specific metrics paired with human or LLM judgments (Xiao et al., 2024; Xu et al., 2025a; Gabriel et al., 2024). Among the 30 papers using LLM judges, 23 reported some form of validation; among 45 papers using rubrics, only 13 reported reliability or agreement (Yang et al., 2025b; Feng et al., 2025). However, evaluator reliability and construct validity are different: agreement between an LLM judge and human raters can establish reproducible scoring without showing that the scoring criteria validly capture the intended construct.

This distinction matters even for established instruments. Scales such as PANAS or WAI, developed for particular human respondents or clinical contexts, can indicate whether a simulated state changes as expected, but they do not provide evidence of corresponding psychological changes in real users. Likewise, study-specific measures of authenticity, resemblance, or profile adherence can support within-study comparison, while differing definitions and scoring schemes make cross-study comparison harder and leave it unclear whether similarly named measures capture the same construct (Qiu et al., 2026; Louie et al., 2024).

The central gap is therefore not whether theory is incorporated into evaluation, but whether the resulting measures validly capture the intended theoretical constructs.

4.2 Extended dialogue is common, but trajectory-level evaluation remains limited

Utility was evaluated in 44 papers, fidelity in 33, realism in 29, emotional plausibility in 28, consistency in 20, and safety in 14; 50 of 52 papers evaluated at least one of these dimensions and 40 evaluated three or more (Appendix Tables 16 and 17). Measures such as empathy, coherence, guidance, authenticity, and therapy-technique use provide important evidence about system quality (Deshpande et al., 2025; Xie et al., 2025; Basar et al., 2025; Xu et al., 2025b), but do not necessarily show how behavior develops across interaction.

Extended dialogue is also common: 31 of 52 papers evaluated conversations of six or more turns, distinct from *Longitudinal* evaluation spanning multiple sessions or a longer period. Of these, the

highest demonstrated behavioral evaluation depth was *Static* in 16, *Pattern-level* in 7, and *Dynamic* in 7. For resistance, Static evaluation can assess whether individual responses appear resistant, and Pattern-level evaluation how often resistance occurs; trajectory-level evaluation asks whether resistance persists, decreases, or re-emerges as the interaction develops. Thus, a longer dialogue does not by itself provide trajectory-level evidence.

More dynamic examples include the Interactive Narrative Therapist, which evaluates therapeutic progression through innovative moments (Feng et al., 2025), and MusPsy, which tracks working alliance and affect across multiple sessions (Wang et al., 2025a). EmoAgent instead measures pre-post changes in simulated psychological assessments (Qiu et al., 2025), capturing endpoint change rather than the intervening behavioral trajectory.

The central gap is therefore not interaction length, but whether persistence or behavioral change is evaluated through evidence about the ordered development of behavior.

4.3 Claim–evidence alignment weakens as conclusions broaden

Across the corpus, papers’ main empirical conclusions were generally well matched to the evidence reported: 41 of 52 (78.8%) were *ALIGNED*, 7 (13.5%) *PARTIAL*, and 4 (7.7%) *EXCEEDS*. Twenty papers also drew a broader conclusion from their findings; among these, 12 (60.0%) were *PARTIAL* and 8 (40.0%) *EXCEEDS*, with none fully aligned (Appendix Table 13). Thus, mismatch was concentrated not in papers’ main empirical conclusions, but when conclusions broadened beyond what the evaluation directly established, for example from response or simulation quality toward training or therapeutic consequences.

This pattern is also reflected in the available human-facing evidence. User studies appeared in 17 papers, lay-user evaluation in 13, and explicit human learning or outcome measures in only 10. Many studies instead relied on model outputs, benchmark performance, simulated dialogue quality, expert ratings, or LLM judges, although a smaller set directly examined human-facing outcomes, including counselor skill development (Louie et al., 2026; Tanana et al., 2019) and perceived empathy or interaction experience (Schmidmaier et al., 2025b,a). Measures of simulated affect or symptom change can provide meaningful evidence about simulated behavior, but

do not establish corresponding effects on real users.

The central issue is therefore not that every interactive system requires human-outcome evaluation. Rather, the scope of the conclusion should remain within the scope of the evidence: response-level evidence supports response-level claims, simulation evidence supports claims about simulated behavior, and stronger claims about therapeutic benefit, or real-world effects require corresponding evidence.

5 Behavior-Grounded Evaluation

The three diagnoses motivate *Behavior-Grounded Evaluation*, a framework that organizes behavioral evaluation as an explicit evidence chain. It is guided by three questions: *What behavior is being measured? How should that behavior develop across interaction? What conclusions can the resulting evidence support?*

We operationalize these questions through seven linked components: *Claim* → *Construct* → *Behavioral Markers* → *Trajectory* → *Intervention* → *Evaluator* → *Evidence Boundary*. These components are not independent evaluation dimensions. Rather, they form an evidence chain: the claim identifies the construct that must be measured; the construct determines the observable markers; the expected trajectory specifies how those markers should evolve; interventions test whether that evolution is sensitive to meaningful interaction; evaluators provide judgments about the resulting evidence; and the evidence boundary specifies which conclusions those judgments can support. Weakness at an earlier link cannot generally be repaired by adding more metrics or evaluators downstream.

5.1 What behavior is being measured?

Evaluation begins by stating the behavioral claim to be evaluated. A behavioral claim should be translated into a *construct* that identifies what property of behavior must be present for the claim to hold. For example, a claim that a simulated client exhibits resistance requires specifying what counts as resistance in the interaction rather than treating any negative or disagreeing response as evidence.

The construct is then translated into observable *behavioral markers*. These markers should specify not only behaviors that count as evidence for the construct, but also nearby behaviors from which the construct must be distinguished. For resistance, markers might include limited disclosure, reluctance to accept suggestions, or sustained dis-

agreement under particular interactional conditions, while general negativity or disengagement should not automatically receive the same interpretation. Defining these markers carefully is important because evaluator agreement does not guarantee construct validity: judges may apply a rubric consistently even when its criteria do not validly capture the intended construct.

These markers also define what an appropriate *evaluator* must be able to judge. Experts, human coders, LLM judges, and automatic metrics provide different forms of evidence and should be validated for the judgments they are asked to make. Evaluator reliability concerns whether judgments can be reproduced, whereas construct validity concerns whether those judgments capture the behavior required by the claim. Reliable scoring therefore does not by itself establish valid measurement.

5.2 How should the behavior develop across interaction?

Once markers are defined, evaluation should specify their expected *trajectory*. A trajectory is an ordered expectation about persistence or change, not simply a sequence of individually acceptable responses. Researchers should specify the relevant interaction window, the markers to be tracked, and whether those markers are expected to persist, increase, decrease, or change following particular interaction events.

For example, an initially guarded simulated client might be expected to remain hesitant after generic encouragement, disclose more after sustained exploration, and become less resistant only after sufficient rapport-building. The relevant test is therefore not whether each response appears plausible in isolation, but whether the sequence of markers follows an interactionally plausible path. This can reveal behavioral drift that aggregate frequency or whole-dialogue ratings conceal.

Interventions provide a diagnostic test of this trajectory. Holding the patient profile and scenario constant while varying a meaningful counselor action allows the expected behavioral consequence to be specified in advance. If sustained exploration is expected to reduce guardedness more than generic encouragement, the evaluation can test whether subsequent markers diverge accordingly. The unit of evaluation is then the relation between an interaction event and the behavior that follows, rather than only the quality of the immediate response.

Intervention-sensitive evaluation is especially

useful for distinguishing genuine interactional adaptation from generic consistency. A system that produces the same plausible behavior regardless of counselor action may appear stable, yet fail the stronger claim that its behavior responds appropriately to interaction.

5.3 What conclusions can the evidence support?

The final component is the *evidence boundary*: the strongest conclusion warranted by the evaluation that has actually been conducted. Different forms of evidence support different scopes of conclusion.

Behavioral markers can show that a system exhibits a target behavior. Trajectory evidence can show that the behavior persists or changes in an expected pattern. Intervention-sensitive evidence can show that the pattern responds appropriately to meaningful differences in interaction. None of these, by themselves, establishes effects on the people who use the system.

The evidence boundary should therefore be stated explicitly in terms of three elements: what the current evaluation supports, what stronger claim it does not yet support, and what additional evidence would be required. For example, a simulated client that exhibits realistic resistance trajectories may support a claim of behavioral validity, while a claim that practicing with the client improves counseling skill additionally requires evidence of trainee learning. Likewise, evidence that an agent avoids unsafe responses in simulated conversations does not by itself establish safety under real-world deployment.

This distinction does not make simulated or model-based evidence uninformative. Rather, it clarifies what conclusions such evidence can support and prevents improvement at one evidential level from being interpreted as progress on a broader human-facing outcome.

6 Case Study: Roleplay-doh

We examine Roleplay-doh (Louie et al., 2024) as a case study of how Behavior-Grounded Evaluation can extend the evaluation of an interactive LLM system. Roleplay-doh enables counseling experts to create simulated patients by translating qualitative feedback into behavioral principles, and evaluates the patients through creator and third-party ratings as well as principle-adherence tests. Table 2 contrasts its existing evaluation with extensions

proposed through Behavior-Grounded Evaluation.

What behavior is being measured? Roleplay-doh provides principles describing behaviors such as skepticism toward suggestions and limited disclosure. Behavior-Grounded Evaluation would use expert judgment to define a candidate construct, such as *guardedness*, and examine whether the proposed markers distinguish it from distrust, anxiety, or disengagement.

How should the behavior develop across interaction? The same markers can then be examined as a trajectory. For example, a guarded patient might remain hesitant after generic encouragement but disclose more gradually as the counselor explores the problem. Operationally, the evaluation could track the occurrence and timing of guardedness-related markers across turns, identify abrupt changes in resistance or disclosure, and compare behavioral trajectories following different counselor actions. Such an analysis could reveal failures that principle-adherence assessments may miss: a patient may appear appropriately resistant at one turn but become implausibly agreeable later, or respond similarly to counselor actions that should have different behavioral consequences.

What conclusions can the evidence support?

Roleplay-doh evaluates whether simulated patients appear authentic, resemble real cases, and are perceived as ready for training. These measures support conclusions about behavioral plausibility and perceived suitability for training, but they do not directly establish that novice counselors improve their skills after using the system. A conclusion about training effectiveness would require evidence of trainee learning or skill development.

Taken together, this case study shows how Behavior-Grounded Evaluation can extend an existing evaluation by making the relationship between constructs and behavioral markers explicit, specifying trajectory- and intervention-sensitive assessments, and clarifying which conclusions the resulting evidence can support.

7 Implications for Behavioral Evaluation

Our review and case study suggest that progress in interactive LLM systems cannot be interpreted through performance scores alone. Its meaning depends on what behavior is being measured, how that behavior develops across interaction, and what conclusions the evidence can support.

More evaluation is not necessarily stronger evaluation. The corpus does not show a field that neglects evaluation. Most papers use several evaluation dimensions, and many combine automatic metrics, experts, rubrics, and LLM judges. The problem is therefore not simply too little evaluation. Adding a metric or evaluator strengthens evidence only when it addresses a missing link between the behavior being evaluated and the conclusion drawn from it. Behavior-Grounded Evaluation shifts attention from the number of evaluation instruments to the structure of the evidence they provide.

Progress is construct-relative. Performance improvements have meaning only relative to the construct the measure is intended to capture. A system can improve in realism, helpfulness, or stylistic similarity without becoming better at maintaining therapeutic stance, modeling resistance, or adapting disclosure appropriately. This matters when benchmarks or LLM judges collapse several desirable properties into a quality score: higher scores may indicate better outputs without establishing improvement in the intended behavioral construct. Claims of progress should therefore identify which construct improved and why the measure provides evidence for that interpretation.

Generalization includes behavior across interaction. For interactive LLM systems, generalization concerns more than performance on unseen prompts, cases, or scenarios. It also concerns whether the behavioral pattern remains appropriate across interaction histories and responds appropriately to user actions. A simulated client may generate plausible responses across cases yet fail behaviorally if resistance disappears abruptly, or disclosure occurs regardless of counselor strategy. Trajectory- and intervention-sensitive evaluation can therefore reveal a form of generalization failure that conventional test-set evaluation may miss.

System progress and human benefit are different evidential targets. Our claim-evidence analysis suggests that inferential risk increases when conclusions extend beyond what the evaluation directly establishes. Behaviorally valid simulation is a research outcome in its own right, but progress in simulation should not automatically be interpreted as progress in training effectiveness, therapeutic benefit, or deployment safety. Explicit evidence boundaries allow researchers to make strong conclusions about system behavior without requiring

Component	Roleplay-doh grounding	Behavior-Grounded extension
Claim	Simulated patients should follow expert-defined principles, remain consistent with their description and context, and resemble real cases.	Separate claims about behavioral simulation from broader conclusions about training effectiveness or novice-counselor outcomes.
Construct	Principles describe hesitation, skepticism toward help, reluctance toward suggestions, and disclosure.	Define a focal construct, such as help-seeking resistance or guardedness, that groups related principles.
Markers	Principles specify observable behaviors such as limited disclosure, skepticism toward suggestions, and hesitation following encouragement.	Treat these as explicit markers of the construct and distinguish them from related behaviors such as negativity or disagreement.
Trajectory	Multi-turn dialogues are collected, and properties such as role consistency are evaluated.	Track how the markers persist or change across turns rather than only whether they occur or whether the dialogue is globally consistent.
Intervention	Some principles are conditional on counselor inputs, and their situational applicability is checked.	Hold the patient profile constant, vary meaningful counselor actions, and pre-specify the expected behavioral consequences.
Evaluator	Creator counselors, third-party counselors, and expert rankings are used.	Use expert-coded markers and, where appropriate, a calibrated LLM judge for scalable marker and trajectory coding.
Evidence boundary	The evaluation measures authenticity, resemblance, principle adherence, role consistency, and perceived readiness for training.	Support conclusions about behavioral simulation and perceived suitability, while reserving conclusions about trainee learning or therapeutic benefit for direct human-outcome evidence.

Table 2: Case study of Behavior-Grounded Evaluation applied to Roleplay-doh, contrasting its reported evaluation with proposed extensions.

every study to conduct a human-outcome evaluation, while making clear when additional human-facing evidence is needed.

Taken together, these implications shift attention from the amount of evaluation to the structure and scope of the evidence. For repurposed interactive LLM systems, progress depends on whether the intended construct is measured validly, whether the target behavior develops appropriately across interaction, and whether conclusions remain within what the evidence can support. Behavior-Grounded Evaluation makes these relationships explicit.

8 Conclusion

Through a structured review of 52 papers, we identified three connected challenges in current evaluation practice: theory grounding is common, but its translation into evaluation measures does not by itself establish construct validity; extended

dialogue often lacks trajectory-level evaluation; and claim–evidence alignment weakens as conclusions broaden beyond what is directly evaluated. Importantly, papers’ main empirical conclusions were usually supported; larger gaps emerged when conclusions extended toward broader behavioral or human-facing consequences. We introduced *Behavior-Grounded Evaluation* to connect behavioral claims with construct measurement, interaction trajectories, interventions, evaluators, and explicit evidence boundaries, and illustrated its use through a case study. For repurposed interactive LLM systems, progress should depend not only on improved outputs, but on whether the intended behavior is measured validly, develops appropriately across interaction, and supports conclusions commensurate with the evidence available.

9 Limitations

This study should be interpreted within the scope of our review design. First, our corpus focuses on LLM-based counseling, therapy, and behavioral simulation. We use this domain as a stress case for examining evaluation practice in repurposed interactive LLM systems, rather than as an exhaustive review of mental-health or interactive LLM applications. Future work should examine whether similar evaluation patterns appear in other human-facing domains, such as education, healthcare communication, or workplace support.

Second, our analysis is based on what papers report. Some studies may have conducted additional validation, prompt testing, or failure analysis that is not fully described in the published text. Our coding therefore captures reported evaluation practice—the evidence available to readers and reviewers.

Third, the claim–evidence analysis focuses on each paper’s main empirical conclusion and, where present, one broader explicit conclusion drawn from the study. It does not enumerate every claim made within each paper. The reported alignment distributions therefore characterize these selected inferential roles rather than the prevalence of unsupported claims across all statements in the literature.

Finally, Behavior-Grounded Evaluation is a claim-matched framework rather than a fixed benchmark. The relevant constructs, behavioral markers, trajectories, interventions, evaluators, and evidence boundaries will vary across domains and system roles. The Roleplay-doh worked application illustrates how the framework can be instantiated, but does not constitute an empirical validation of the framework itself. Future work should test its usefulness and reliability across systems and domains.

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A Reviewed Corpus

Tables 3–5 list the 52 papers included in the review.

B Coding Scheme

The codebook contains 40 coding fields organized into seven groups: system and simulation design, evaluation methods, evaluation dimensions, interaction and behavioral evaluation, theory, transparency and reliability, additional evaluation characteristics, and claim–evidence alignment. Bibliographic metadata and traceability information, such as claim/evidence quotations and page locations, were recorded separately and are not counted among the 40 coding fields. Tables 6–12 provide the field definitions, allowable values, and coding rules.

B.1 System and Simulation Design

These fields characterize the primary role and design of the simulated system, including what is simulated, how personas or behavioral states are represented, and whether those states change over interaction. Table 6 summarizes the corresponding fields and coding rules.

B.2 Evaluation Methods

These fields record who or what performs the evaluation, distinguishing domain experts, lay users, interactive user studies, LLM judges, and automatic metrics. Table 7 provides the definitions and coding rules.

B.3 Evaluation Dimensions

These fields capture what aspects of system performance are explicitly evaluated, including safety, realism, consistency, fidelity, human outcomes, utility, and emotional plausibility. Table 8 also includes the free-text field used to record reported evaluation metrics.

B.4 Interaction and Behavioral Evaluation

These fields characterize how much interaction is represented in the evaluation, how deeply behavior is analyzed across interaction, and whether behavior is tested under different input or intervention conditions. Table 9 defines these fields and their coding categories.

B.5 Theory, Transparency, and Reliability

These fields capture theory grounding and operationalization, the types of theory-related evaluation

used, prompt disclosure, clinical theory, and reporting of evaluator agreement or reliability. Theory grounding captures whether established theory substantively informs the system or study, whereas theory operationalization captures whether that theory is translated into measurable evaluation constructs, criteria, or procedures. Table 10 summarizes the corresponding definitions and coding rules.

B.6 Additional Evaluation Characteristics

These fields capture additional characteristics of evaluation practice and transparency, including rubric use, LLM-judge validation, comparability, longitudinal and robustness testing, failure analysis, simulation realism, and dataset availability. Table 11 provides the corresponding coding rules.

B.7 Claim–Evidence Alignment

The final two fields capture the relationship between the conclusions drawn in a paper and the evidence reported to support them. Unlike the other paper-level fields, these fields were applied to predefined claim roles: the primary empirical or system-level claim (C1) and, where present, an explicit higher-scope inferential extension beyond C1 (C2).

Claims were selected primarily from the abstract, explicit contribution statements, and conclusion; discussion sections were used only when they contained an explicit, nonredundant empirical conclusion. C2 was identified when a paper drew an explicit conclusion extending beyond its primary empirical conclusion (C1), regardless of whether the available evidence supported that extension. We distinguished such conclusions from speculative future applications: statements describing potential uses, research objectives, or outcomes requiring future investigation were excluded. Hedged language alone did not determine eligibility; the key criterion was whether the statement was presented as an inference from the reported findings rather than a prospective possibility. Importantly, a C2 claim could be classified as ALIGNED if the reported evidence directly supported its broader conclusion.

For each selected claim, the full paper was examined to identify the strongest directly relevant supporting evidence. Alignment judgments followed an evidence-level preservation principle: evidence at the response, simulation, perception, behavioral, or human-outcome level was treated as support for claims at that evidential level, rather than automatically as support for stronger down-

No.	Citation and title	Venue
1	Xiao et al., 2024. <i>HealMe: Harnessing Cognitive Reframing in Large Language Models for Psychotherapy</i>	ACL
2	Gabriel et al., 2024. <i>Can AI Relate: Testing Large Language Model Response for Mental Health Support</i>	EMNLP
3	Haydarov et al., 2025. <i>Towards AI-Assisted Psychotherapy: Emotion-Guided Generative Interventions</i>	EMNLP
4	Qiu et al., 2025. <i>EmoAgent: Assessing and Safeguarding Human-AI Interaction for Mental Health Safety</i>	EMNLP
5	Aghakhani et al., 2025. <i>From Conversation to Automation: Leveraging LLMs for Problem-Solving Therapy Analysis</i>	ACL
6	Wang et al., 2025c. <i>AnnaAgent: Dynamic Evolution Agent System with Multi-Session Memory for Realistic Seeker Simulation</i>	ACL
7	Wang et al., 2026. <i>CARE-Bench: A Benchmark of Diverse Client Simulations Guided by Expert Principles for Evaluating LLMs in Psychological Counseling</i>	AAAI
8	Sungarda et al., 2025. <i>GenAI for Social Work Field Education: Client Simulation with Real-Time Feedback</i>	IEEE BigData
9	Liu et al., 2023. <i>ChatCounselor: A Large Language Models for Mental Health Support</i>	CIKM 2023
10	Chiu et al., 2024. <i>A Computational Framework for Behavioral Assessment of LLM Therapists</i>	arXiv
11	Li et al., 2024. <i>Understanding the Therapeutic Relationship between Counselors and Clients in Online Text-based Counseling using LLMs</i>	arXiv
12	Wang et al., 2024a. <i>Towards a Client-Centered Assessment of LLM Therapists by Client Simulation</i>	arXiv
13	Hu et al., 2025. <i>PsycoLLM: Enhancing LLM for Psychological Understanding and Evaluation</i>	IEEE
14	Fouda et al., 2026. <i>PsychiatryBench: a multi-task benchmark for LLMs in psychiatry</i>	npj Digital Medicine
15	Cai et al., 2026. <i>Exploring Safety Alignment Evaluation of LLMs in Chinese Mental Health Dialogues via LLM-as-Judge</i>	arXiv
16	Qiu et al., 2026. <i>PsyCLIENT: Client Simulation via Conversational Trajectory Modeling for Trainee Practice and Model Evaluation in Mental Health Counseling</i>	arXiv
17	Nguyen et al., 2025. <i>Do Large Language Models Align with Core Mental Health Counseling Competencies?</i>	NAACL 2025
18	Zhang et al., 2025a. <i>CBT-BENCH: Evaluating Large Language Models on Assisting Cognitive Behavior Therapy</i>	ACL

Table 3: Papers included in the 52-paper review corpus (Part 1 of 3).

No.	Citation and title	Venue
19	Kim et al., 2025a . <i>KMI: A Dataset of Korean Motivational Interviewing Dialogues for Psychotherapy</i>	ACL
20	Deshpande et al., 2025 . <i>Evaluating Large Language Models for Enhancing Live Chat Therapy: A Comparative Study with Psychotherapists</i>	ACL
21	Jaiswal et al., 2024 . <i>Building Personality-Adaptive Conversational AI for Mental Health Therapy</i>	ACM BCB '24
22	Xu et al., 2025a . <i>MentalChat16K: A Benchmark Dataset for Conversational Mental Health Assistance</i>	ACM KDD'25
23	Schmidmaier et al., 2025b . <i>Using a Secondary Channel to Display the Internal Empathic Resonance of LLM-Driven Agents for Mental Health Support</i>	ACM ICMIMLM'25
24	Schmidmaier et al., 2025a . <i>Using Nonverbal Cues in Empathic Multi-Modal LLM-Driven Chatbots for Mental Health Support</i>	ACM
25	Ozgun et al., 2025 . <i>Trustworthy AI Psychotherapy: Multi-Agent LLM Workflow for Counseling and Explainable Mental Disorder Diagnosis</i>	ACM CIKM '25
26	Louie et al., 2026 . <i>Can LLM-Simulated Practice and Feedback Upskill Human Counselors? A Randomized Study with 90+ Novice Counselors</i>	ACM CHI '26
27	Kuhail et al., 2025 . <i>Human-Human vs Human-AI Therapy: An Empirical Study</i>	IJHCI
28	Louie et al., 2024 . <i>Roleplay-doh: Enabling Domain-Experts to Create LLM-simulated Patients via Eliciting and Adhering to Principles</i>	EMNLP
29	Wang et al., 2024b . <i>PATIENT-Ψ: Using Large Language Models to Simulate Patients for Training Mental Health Professionals</i>	EMNLP
30	Lee et al., 2024 . <i>CACTUS: Towards Psychological Counseling Conversations using Cognitive Behavioral Theory</i>	EMNLP
31	Basar et al., 2024 . <i>To What Extent Are Large Language Models Capable of Generating Substantial Reflections for Motivational Interviewing Counseling Chatbots? A Human Evaluation</i>	ACL
32	Yang et al., 2025b . <i>Consistent Client Simulation for Motivational Interviewing-based Counseling</i>	ACL
33	Yang et al., 2025a . <i>CAMI: A Counselor Agent Supporting Motivational Interviewing through State Inference and Topic Exploration</i>	ACL
34	Xie et al., 2025 . <i>PsyDT: Using LLMs to Construct the Digital Twin of Psychological Counselor with Personalized Counseling Style for Psychological Counseling</i>	ACL
35	Kermani et al., 2025 . <i>A Systematic Evaluation of LLM Strategies for Mental Health Text Analysis: Fine-tuning vs. Prompt Engineering vs. RAG</i>	ACL 2025 CLPsych

Table 4: Papers included in the 52-paper review corpus (Part 2 of 3).

No.	Citation and title	Venue
36	Basar et al., 2025 . <i>How Well Can Large Language Models Reflect? A Human Evaluation of LLM-generated Reflections for Motivational Interviewing Dialogues</i>	ACL COLING
37	Feng et al., 2025 . <i>Reframe Your Life Story: Interactive Narrative Therapist and InNovative Moment Assessment with Large Language Models</i>	EMNLP 2025
38	Xu et al., 2025b . <i>MultiAgentESC: A LLM-based Multi-Agent Collaboration Framework for Emotional Support Conversation</i>	EMNLP 2025
39	Kim et al., 2025c . <i>MIRROR: Multimodal Cognitive Reframing Therapy for Rolling with Resistance</i>	EMNLP 2025
40	Kim et al., 2025b . <i>Can You Share Your Story? Modeling Clients' Metacognition and Openness for LLM Therapist Evaluation</i>	ACL 2025
41	Liu et al., 2025 . <i>Eeyore: Realistic Depression Simulation via Expert-in-the-Loop Supervised and Preference Optimization</i>	ACL
42	Wang et al., 2025d . <i>Feel the Difference? A Comparative Analysis of Emotional Arcs in Real and LLM-Generated CBT Sessions</i>	EMNLP
43	Samuel et al., 2025 . <i>PersonaGym: Evaluating Persona Agents and LLMs</i>	EMNLP
44	Chen et al., 2025 . <i>MIND: Towards Immersive Psychological Healing with Multi-Agent Inner Dialogue</i>	EMNLP
45	Berrezueta-Guzman et al., 2024 . <i>Future of ADHD Care: Evaluating the Efficacy of ChatGPT in Therapy Enhancement</i>	MDPI(Healthcare)
46	Na, 2024 . <i>CBT-LLM: A Chinese Large Language Model for Cognitive Behavioral Therapy-based Mental Health Question Answering</i>	ACL
47	Sharma et al., 2023 . <i>Cognitive Reframing of Negative Thoughts through Human-Language Model Interaction</i>	ACL
48	Kang et al., 2024 . <i>Can Large Language Models be Good Emotional Supporter? Mitigating Preference Bias on Emotional Support Conversation</i>	ACL
49	Chen et al., 2023 . <i>LLM-empowered Chatbots for Psychiatrist and Patient Simulation: Application and Evaluation</i>	arXiv
50	Schmidgall et al., 2025 . <i>AgentClinic: a multimodal agent benchmark to evaluate AI in simulated clinical environments</i>	arXiv
51	Wang et al., 2025a . <i>Psychological Counseling CanNot Be Achieved Overnight: Automated Psychological Counseling Through Multi-Session Conversations</i>	arXiv
52	Tanana et al., 2019 . <i>Development and Evaluation of ClientBot: Patient-Like Conversational Agent to Train Basic Counseling Skills</i>	JMIR

Table 5: Papers included in the 52-paper review corpus (Part 3 of 3).

Field	Values / Type	Definition and coding rule
Focus_Type	Open classification	Primary research focus of the paper. Examples in the codebook include Simulation Framework, Evaluation Framework, Dataset, Training System, and Tool.
Simulation_Target	Client Agent / Therapist Agent / Dual-Agent / Human Trainee	Object or participant role simulated by the system.
Persona_Model_Depth	Surface Persona / Behavioral State Model / Cognitive Model / Dynamic Traits / Expert Principles	Depth of the simulated user or client model. Surface Persona uses profile-level descriptions; Behavioral State Model represents behavioral states; Cognitive Model includes beliefs or motivations; Dynamic Traits change over interaction; Expert Principles use expert-defined behavioral principles.
Uses_Dynamic_State	Yes / No	Whether the system explicitly tracks and updates a psychological or user state, such as state transitions, session memory, dynamic openness, or related mechanisms.
Temporal_Modeling_Details	Free text	Description of how dynamic or temporal state is modeled, such as state transitions, session memory, dynamic openness, or updates to conversational principles.

Table 6: Coding fields for system and simulation design.

stream conclusions. Claim and evidence quotations and their page or section locations were retained for traceability.

Operationally, we distinguished PARTIAL from EXCEEDS by asking whether the claim could be narrowed to match the reported evidence without changing its fundamental construct, outcome, population, or evidential level. If so, it was coded PARTIAL; if narrowing would change the fundamental target of the claim, it was coded EXCEEDS.

Table 13 illustrates how these rules were applied to claims in the reviewed corpus.

C Coding Reliability

For the 40 fields, two researchers independently coded the full corpus before adjudication. Mean raw agreement across fields was 82.5%; among fields for which Cohen's κ was computable, mean κ was 0.61. Disagreements and ambiguous cases were resolved through discussion with the faculty lead.

D Supplementary Results

Field	Values / Type	Definition and coding rule
Eval_Human_Experts	Yes / No	Evaluation by domain experts such as therapists, clinicians, domain specialists, or trained annotators. Excludes crowd workers, participants without domain expertise, and LLM evaluators.
Eval_Lay_Users	Yes / No	Evaluation by ordinary users or participants without relevant domain expertise, including crowd workers or participants whose professional expertise is not specified.
Eval_User_Study	Yes / No	Participants actively interact with the system under a defined task, protocol, or experimental design. Static output-rating tasks without interaction are excluded.
Eval_LLM_Judge	Yes / No	An LLM such as GPT-4 or Claude is prompted to evaluate, score, compare, or rank system outputs. Embedding similarity, feature extraction, or algorithmic metrics alone do not count.
Eval_Automatic	Yes / No	Evaluation calculated automatically through an algorithm or formula, such as BLEU, ROUGE, Accuracy, F1, cosine similarity, or task success rate. Human and LLM judgments are excluded.

Table 7: Coding fields for evaluation methods.

Field			Values/Type	Definition and coding rule
Safety			Yes / No	Whether the evaluation assesses harmful, biased, unsafe, or ethically problematic outputs. General usefulness alone does not count.
Realism			Yes / No	Whether outputs or behaviors are evaluated as realistic, human-like, natural, or believable. Correctness or task success alone does not count.
Consistency			Yes / No	Whether behavior is evaluated for stability across turns, such as persona consistency, behavioral drift, or trait alignment. Single-turn quality alone does not count.
Fidelity			Yes / No	Whether outputs conform to a predefined identity, profile, role, or target model. General realism alone does not count.
Human Outcomes	Learning	/	Yes / No	Whether evaluation measures substantive change in human knowledge, skills, or behavior. Perception-only measures are excluded.
Utility			Yes / No	Whether the system is evaluated for usefulness, effectiveness, helpfulness, or task performance. Realism alone does not count.
Emotional Plausibility			Yes / No	Whether emotional behavior or empathic responses are evaluated as realistic or appropriate. Factual correctness alone does not count.
Raw Eval Metrics			Free text	Evaluation metrics reported by the paper, recorded verbatim or in normalized form for descriptive analysis.

Table 8: Coding fields for evaluation dimensions and reported metrics.

Field	Values/Type	Definition and coding rule
Interaction_Level	Single-turn / Short Multi-turn / Extended Dialogue / Longitudinal	Amount of interaction represented in the evaluation. Single-turn evaluates one response without dialogue history; Short Multi-turn contains 2–5 turns; Extended Dialogue uses longer conversations, typically 6 or more turns; Longitudinal evaluation spans multiple sessions or a longer period. When multiple interaction levels were present, the highest level demonstrated in the evaluation was coded. The six-turn threshold is a pragmatic coding convention used to distinguish short from extended within-session interactions, rather than a theoretical boundary for behavioral change.
Behavior_Eval_Depth	Dynamic / Pattern-level / Static / None	Depth of behavioral evaluation. Dynamic examines how behavior develops across ordered observations within an interaction, such as turn-by-turn trends, transitions between dialogue phases, or systematic qualitative analysis of behavioral progression. Pattern-level summarizes behavioral frequencies or patterns without tracking change; Static scores behavior independently without examining cross-turn relations; None evaluates output quality without behavioral analysis. Independent response ratings, aggregate behavioral frequencies, and pre–post endpoint comparisons without analysis of the intervening interaction do not by themselves qualify as Dynamic. When multiple forms of behavioral evaluation were present, the highest demonstrated evaluation depth was coded.
Intervention_Sensitivity	Yes / No	Whether the paper explicitly tests differences in system behavior under different inputs or intervention conditions. Papers that do not test or report such behavioral differences are coded No.

Table 9: Coding fields for interaction structure and behavioral evaluation.

Field	Values / Type	Definition and coding rule
Theory_Operationalized	Strong / Partial / None	Degree to which theory is translated into measurable constructs, rubrics, behavioral criteria, or evaluation procedures. Strong indicates that theory explicitly defines substantial parts of the evaluation; Partial indicates that theory is translated into evaluation criteria only incompletely or for a subset of the theoretically relevant constructs; None indicates that theory may inform the system or study but is not operationalized in the evaluation.
Prompt_Disclosure	Full / Partial / No Disclosure	Extent to which prompts needed to understand or reproduce the system and evaluation are reported. Full Disclosure provides the necessary prompts; Partial Disclosure provides incomplete prompts or examples; No Disclosure provides no usable prompt information.
Theory_Grounding	Strong / Partial / Mentioned	Extent to which established psychological, clinical, counseling, or related theory substantively informs the system or study. Strong indicates that theory clearly shapes the system design, target constructs, or study rationale; Partial indicates that theory informs the work but only incompletely or indirectly; Mentioned indicates that theory appears mainly as background or motivation without substantively shaping the system or study.
Clinical_Theory	Free text	Specific psychological, clinical, or counseling theory or framework identified in the paper, such as Cognitive Behavioral Therapy or Motivational Interviewing.
Reliability_Reported	Yes / No / N/A	Whether evaluator agreement or inter-rater reliability is explicitly reported. Yes indicates a reported agreement statistic or percentage; No indicates human evaluation without reported agreement; N/A applies when no human evaluation is used.
Agreement Method	Free text	Specific agreement or reliability method reported in the paper, such as Cohen's kappa, Krippendorff's alpha, Fleiss' kappa, or percentage agreement. Left blank when no method is reported.
Theory_Eval_Type	Multi-label categorical	Type or types of theory-related evaluation used in the paper. Categories are not mutually exclusive; the evaluation types relevant to the reported analyses are summarized in Table 21.

Table 10: Coding fields for theory grounding, theory-related evaluation, transparency, and reliability.

Field	Values/Type	Definition and coding rule
Has_Rubric	Yes / No	Whether the evaluation provides explicit scoring criteria, a rating scale, rubric, or scoring guidance. Merely stating that annotators provided ratings without criteria is coded No.
LLM_Judge_Validated	Yes / No	Whether LLM-judge scores are compared with human judgments in any form, including correlation, agreement tests, mean comparisons, inter-rater tables, or qualitative statements of agreement. Human-human reliability alone does not count. This field records whether any comparison with human judgments was reported and does not imply formal psychometric validation.
Uses_Standard_Metrics	Yes / No	Whether established evaluation metrics such as BLEU, ROUGE, or BERTScore are used. Papers using only custom metrics without comparison to standard metrics are coded No.
Metric_Interpretable	Yes / No	Whether the meaning of the evaluation metric and the interpretation of its scores are clear to the reader.
Comparable_To_Prior_Work	Yes / No	Whether the evaluation uses metrics or datasets that permit comparison with prior work. A custom dataset combined with custom metrics and no baseline is coded No.
Has_Longitudinal_Eval	Yes / No	Whether evaluation spans multiple sessions or a longer period. A single-session evaluation is coded No.
Has_Robustness_Testing	Yes / No	Whether the system is tested under multiple conditions or datasets rather than only under ideal conditions.
Has_Failure_Analysis	Yes / No	Whether the paper explicitly analyzes failure cases, for example in a dedicated failure-analysis section or table.
Sim_Behavior_Realistic	Yes / No	Whether simulated behavior includes realistic characteristics such as resistance or emotional fluctuation rather than being overly compliant or mechanical.
Dataset_Available	Yes / No	Whether the dataset is publicly available through a download link or repository.

Table 11: Coding fields for additional evaluation characteristics.

Field	Values / Type	Definition and coding rule
Claim	C1 / C2; free text	Selected empirical claim and its inferential role. C1 captures the paper’s primary empirical or system-level conclusion. C2 is optional and captures the strongest explicit, non-speculative inferential extension beyond C1. Motivational statements, design goals, hypotheses, recommendations, future work, and purely speculative statements are excluded.
Claim_Evidence_Alignment	Aligned / Partial / Exceeds / Unclear	Degree to which the strongest reported evidence supports the selected claim. ALIGNED indicates that the evidence directly evaluates the claimed target at the claimed scope and evidential level. PARTIAL indicates directly relevant evidence for the same fundamental target, but with narrower scope, population, duration, or outcome. EXCEEDS indicates that the claim extends to a materially different construct, population, outcome, or evidential level. UNCLEAR indicates insufficient or contradictory reporting for a reliable judgment.

Table 12: Coding fields for claim–evidence alignment.

Paper / role	Selected claim	Strongest supporting evidence	Alignment
Liu et al., 2025 , C1	Eeyore outperforms GPT-4o-based simulation baselines in linguistic authenticity and profile adherence.	Fifteen counselors or advanced psychology students evaluated live interactions on unseen profiles for authenticity and profile adherence (Sec. 4.2; Fig. 4). The evaluated outcomes therefore directly correspond to the comparative claim.	ALIGNED
Sungarda et al., 2025 , C2	SWITCH provides a scalable, low-cost, and consistent training workflow that can complement field education.	The system was introduced in a compulsory social-work practice course and teacher workshops, supporting its use as a complementary training workflow; however, cost, scalability, and consistency were not directly measured (Sec. VI; Tables VI–VII).	PARTIAL
Chen et al., 2025 , C2	MIND’s inner-dialogue interactions enhance empathy and emotional resonance, reduce self-criticism, and strengthen self-identity.	A short-term study with eight participants measured affect using PANAS and subjective user-experience ratings, together with expert ratings of generated content; self-criticism and self-identity were not directly measured (Sec. 3.4; Tables 3–4).	EXCEEDS

Table 13: Illustrative applications of the claim–evidence alignment coding. The examples show how alignment depends on whether the reported evidence preserves the target and evidential scope of the selected claim. They are illustrative rather than exhaustive; complete claim–evidence records are included in the released coding materials.

Evaluator Type	Count (%)
Human Experts	34 (65.4%)
LLM Judges	30 (57.7%)
Automatic Metrics	35 (67.3%)
All Three (Expert + LLM + Auto)	15 (28.8%)

Table 14: Distribution of evaluator types across reviewed studies ($N = 52$). Counts indicate the number of papers using each evaluator type; categories are not mutually exclusive, except for the final row, which reports papers combining human experts, LLM judges, and automatic metrics.

Evaluator Type	Papers
Human Experts	Aghakhani et al., 2025, Berrezueta-Guzman et al., 2024, Cai et al., 2026, Chen et al., 2023, Chen et al., 2025, Deshpande et al., 2025, Feng et al., 2025, Gabriel et al., 2024, Haydarov et al., 2025, Kim et al., 2025b, Kim et al., 2025a, Kim et al., 2025c, Kuhail et al., 2025, Lee et al., 2024, Li et al., 2024, Liu et al., 2025, Louie et al., 2024, Louie et al., 2026, Na, 2024, Nguyen et al., 2025, Qiu et al., 2026, Samuel et al., 2025, Schmidgall et al., 2025, Sharma et al., 2023, Wang et al., 2024b, Wang et al., 2026, Wang et al., 2025a, Xiao et al., 2024, Xie et al., 2025, Xu et al., 2025a, Xu et al., 2025b, Yang et al., 2025a, Yang et al., 2025b, Zhang et al., 2025a
LLM Judges	Aghakhani et al., 2025, Cai et al., 2026, Chiu et al., 2024, Feng et al., 2025, Fouda et al., 2026, Haydarov et al., 2025, Kim et al., 2025b, Kim et al., 2025c, Lee et al., 2024, Li et al., 2024, Liu et al., 2023, Liu et al., 2025, Louie et al., 2026, Nguyen et al., 2025, Ozgun et al., 2025, Qiu et al., 2025, Qiu et al., 2026, Samuel et al., 2025, Schmidgall et al., 2025, Wang et al., 2024b, Wang et al., 2024a, Wang et al., 2025c, Wang et al., 2026, Wang et al., 2025a, Xiao et al., 2024, Xie et al., 2025, Xu et al., 2025a, Xu et al., 2025b, Yang et al., 2025a, Yang et al., 2025b
Automatic Metrics	Aghakhani et al., 2025, Basar et al., 2025, Chen et al., 2023, Chiu et al., 2024, Deshpande et al., 2025, Feng et al., 2025, Fouda et al., 2026, Gabriel et al., 2024, Hu et al., 2025, Kang et al., 2024, Kermani et al., 2025, Kim et al., 2025b, Li et al., 2024, Louie et al., 2026, Na, 2024, Nguyen et al., 2025, Ozgun et al., 2025, Qiu et al., 2025, Samuel et al., 2025, Schmidgall et al., 2025, Schmidmaier et al., 2025a, Sharma et al., 2023, Sungarda et al., 2025, Tanana et al., 2019, Wang et al., 2024b, Wang et al., 2024a, Wang et al., 2025c, Wang et al., 2026, Wang et al., 2025d, Wang et al., 2025a, Xie et al., 2025, Xu et al., 2025b, Yang et al., 2025a, Yang et al., 2025b, Zhang et al., 2025a
All Three (Expert + LLM + Auto)	Aghakhani et al., 2025, Feng et al., 2025, Kim et al., 2025b, Li et al., 2024, Louie et al., 2026, Nguyen et al., 2025, Samuel et al., 2025, Schmidgall et al., 2025, Wang et al., 2024b, Wang et al., 2026, Wang et al., 2025a, Xie et al., 2025, Xu et al., 2025b, Yang et al., 2025a, Yang et al., 2025b

Table 15: Evaluator types used in reviewed LLM-based counseling and role-play simulation papers. Papers are grouped by whether evaluation involved human experts, LLM judges, automatic metrics, or all three evaluator types.

Evaluation Dimension	Count (%)
Utility	44 (84.6%)
Fidelity	33 (63.5%)
Realism	29 (55.8%)
Emotional Plausibility	28 (53.8%)
Consistency	20 (38.5%)
Safety	14 (26.9%)
Human Learning / Outcomes	10 (19.2%)

Table 16: Evaluation dimensions reported across reviewed studies ($N = 52$). Utility, fidelity, realism, and emotional plausibility were more commonly evaluated than consistency, safety, and human learning or outcome measures; categories are not mutually exclusive.

Category	Papers
Utility	Aghakhani et al., 2025, Basar et al., 2024, Basar et al., 2025, Berrezueta-Guzman et al., 2024, Cai et al., 2026, Chen et al., 2023, Chen et al., 2025, Deshpande et al., 2025, Feng et al., 2025, Fouda et al., 2026, Gabriel et al., 2024, Haydarov et al., 2025, Hu et al., 2025, Jaiswal et al., 2024, Kang et al., 2024, Kermani et al., 2025, Kim et al., 2025b, Kim et al., 2025a, Kim et al., 2025c, Kuhail et al., 2025, Lee et al., 2024, Li et al., 2024, Liu et al., 2023, Louie et al., 2024, Louie et al., 2026, Na, 2024, Nguyen et al., 2025, Ozgun et al., 2025, Qiu et al., 2025, Qiu et al., 2026, Schmidgall et al., 2025, Schmidmaier et al., 2025b, Sharma et al., 2023, Tanana et al., 2019, Wang et al., 2024b, Wang et al., 2024a, Wang et al., 2026, Wang et al., 2025a, Xiao et al., 2024, Xie et al., 2025, Xu et al., 2025a, Xu et al., 2025b, Yang et al., 2025a, Zhang et al., 2025a
Fidelity	Chen et al., 2023, Chen et al., 2025, Chiu et al., 2024, Deshpande et al., 2025, Feng et al., 2025, Fouda et al., 2026, Gabriel et al., 2024, Kang et al., 2024, Kim et al., 2025b, Kim et al., 2025a, Kim et al., 2025c, Li et al., 2024, Liu et al., 2023, Liu et al., 2025, Louie et al., 2024, Louie et al., 2026, Na, 2024, Ozgun et al., 2025, Qiu et al., 2026, Samuel et al., 2025, Schmidgall et al., 2025, Tanana et al., 2019, Wang et al., 2024b, Wang et al., 2024a, Wang et al., 2025c, Wang et al., 2026, Wang et al., 2025d, Wang et al., 2025a, Xiao et al., 2024, Xie et al., 2025, Yang et al., 2025a, Yang et al., 2025b, Zhang et al., 2025a
Realism	Basar et al., 2024, Basar et al., 2025, Berrezueta-Guzman et al., 2024, Chen et al., 2023, Chen et al., 2025, Chiu et al., 2024, Deshpande et al., 2025, Kim et al., 2025b, Kim et al., 2025a, Kuhail et al., 2025, Lee et al., 2024, Liu et al., 2023, Liu et al., 2025, Louie et al., 2024, Louie et al., 2026, Ozgun et al., 2025, Qiu et al., 2026, Samuel et al., 2025, Schmidgall et al., 2025, Tanana et al., 2019, Wang et al., 2024b, Wang et al., 2024a, Wang et al., 2025c, Wang et al., 2026, Wang et al., 2025d, Xie et al., 2025, Xu et al., 2025b, Yang et al., 2025a, Yang et al., 2025b
Emotional Plausibility	Berrezueta-Guzman et al., 2024, Cai et al., 2026, Chen et al., 2023, Chen et al., 2025, Gabriel et al., 2024, Haydarov et al., 2025, Kang et al., 2024, Kim et al., 2025b, Kim et al., 2025c, Kuhail et al., 2025, Liu et al., 2023, Liu et al., 2025, Ozgun et al., 2025, Qiu et al., 2026, Schmidgall et al., 2025, Schmidmaier et al., 2025b, Sharma et al., 2023, Wang et al., 2024b, Wang et al., 2024a, Wang et al., 2026, Wang et al., 2025d, Wang et al., 2025a, Xiao et al., 2024, Xie et al., 2025, Xu et al., 2025a, Xu et al., 2025b, Yang et al., 2025a, Zhang et al., 2025a
Consistency	Berrezueta-Guzman et al., 2024, Chen et al., 2023, Chen et al., 2025, Chiu et al., 2024, Fouda et al., 2026, Gabriel et al., 2024, Haydarov et al., 2025, Kim et al., 2025a, Li et al., 2024, Liu et al., 2025, Louie et al., 2024, Ozgun et al., 2025, Qiu et al., 2026, Wang et al., 2024a, Wang et al., 2025c, Wang et al., 2026, Wang et al., 2025a, Xiao et al., 2024, Yang et al., 2025a, Yang et al., 2025b
Safety	Basar et al., 2024, Basar et al., 2025, Berrezueta-Guzman et al., 2024, Cai et al., 2026, Fouda et al., 2026, Gabriel et al., 2024, Li et al., 2024, Qiu et al., 2025, Samuel et al., 2025, Sharma et al., 2023, Wang et al., 2024a, Xiao et al., 2024, Xie et al., 2025, Xu et al., 2025a
Human Learning / Outcomes	Berrezueta-Guzman et al., 2024, Chen et al., 2025, Feng et al., 2025, Li et al., 2024, Louie et al., 2026, Qiu et al., 2026, Tanana et al., 2019, Wang et al., 2024b, Wang et al., 2025a, Xiao et al., 2024

Table 17: Reviewed papers by evaluation dimension. Papers may appear in multiple rows because evaluation dimensions are not mutually exclusive.

Category	Strong	Partial	Mentioned	None	Total
Theory Grounding	40 (76.9%)	10 (19.2%)	2 (3.8%)	—	52
Theory Operationalization	30 (57.7%)	16 (30.8%)	—	6 (11.5%)	52

Table 18: Distribution of theory grounding and theory operationalization across reviewed studies. Theory grounding captures whether established theory substantively informs the system or study, whereas theory operationalization captures whether theory is translated into measurable constructs, rubrics, behavioral criteria, or evaluation procedures.

Category	Papers
Strong	Xiao et al., 2024, Gabriel et al., 2024, Haydarov et al., 2025, Qiu et al., 2025, Aghakhani et al., 2025, Wang et al., 2026, Sungarda et al., 2025, Chiu et al., 2024, Li et al., 2024, Wang et al., 2024a, Fouda et al., 2026, Cai et al., 2026, Nguyen et al., 2025, Zhang et al., 2025a, Kim et al., 2025a, Jaiswal et al., 2024, Xu et al., 2025a, Schmidmaier et al., 2025b, Schmidmaier et al., 2025a, Ozgun et al., 2025, Louie et al., 2026, Wang et al., 2024b, Lee et al., 2024, Basar et al., 2024, Yang et al., 2025b, Yang et al., 2025a, Xie et al., 2025, Basar et al., 2025, Feng et al., 2025, Xu et al., 2025b, Kim et al., 2025c, Kim et al., 2025b, Liu et al., 2025, Wang et al., 2025d, Samuel et al., 2025, Chen et al., 2025, Na, 2024, Sharma et al., 2023, Wang et al., 2025a, Tanana et al., 2019
Partial	Wang et al., 2025c, Liu et al., 2023, Hu et al., 2025, Qiu et al., 2026, Deshpande et al., 2025, Louie et al., 2024, Kermani et al., 2025, Kang et al., 2024, Chen et al., 2023, Schmidgall et al., 2025
Mentioned	Kuhail et al., 2025, Berrezueta-Guzman et al., 2024

Table 19: Reviewed papers classified by theory grounding level. Papers are grouped according to whether theory grounding was coded as strong, partial, or mentioned only.

Category	Papers
Strong	Xiao et al., 2024, Gabriel et al., 2024, Qiu et al., 2025, Aghakhani et al., 2025, Wang et al., 2026, Sungarda et al., 2025, Chiu et al., 2024, Li et al., 2024, Wang et al., 2024a, Cai et al., 2026, Nguyen et al., 2025, Zhang et al., 2025a, Kim et al., 2025a, Xu et al., 2025a, Schmidmaier et al., 2025b, Schmidmaier et al., 2025a, Louie et al., 2026, Wang et al., 2024b, Lee et al., 2024, Yang et al., 2025a, Feng et al., 2025, Kim et al., 2025c, Kim et al., 2025b, Wang et al., 2025d, Samuel et al., 2025, Na, 2024, Sharma et al., 2023, Kang et al., 2024, Wang et al., 2025a, Tanana et al., 2019
Partial	Haydarov et al., 2025, Liu et al., 2023, Hu et al., 2025, Fouda et al., 2026, Qiu et al., 2026, Jaiswal et al., 2024, Ozgun et al., 2025, Louie et al., 2024, Basar et al., 2024, Yang et al., 2025b, Kermani et al., 2025, Basar et al., 2025, Liu et al., 2025, Chen et al., 2025, Chen et al., 2023, Schmidgall et al., 2025
None	Wang et al., 2025c, Deshpande et al., 2025, Kuhail et al., 2025, Xie et al., 2025, Xu et al., 2025b, Berrezueta-Guzman et al., 2024

Table 20: Reviewed papers grouped by theory operationalization level. Operationalization refers to whether theory is translated into measurable constructs, rubrics, behavioral criteria, or evaluation procedures.

Evaluation Type	Papers
Validated scale <i>Standardized psychological or clinical scales</i>	Xiao et al., 2024, Qiu et al., 2025, Li et al., 2024, Wang et al., 2024a, Schmidmaier et al., 2025b, Schmidmaier et al., 2025a, Yang et al., 2025a, Chen et al., 2025, Wang et al., 2025a
Established therapy/counseling coding system <i>Therapy process or outcome coding</i>	Xiao et al., 2024, Zhang et al., 2025a, Lee et al., 2024, Feng et al., 2025, Kim et al., 2025c, Kim et al., 2025b, Tanana et al., 2019
Theory-specific rubric <i>Custom rubrics based on theoretical constructs</i>	Xiao et al., 2024, Gabriel et al., 2024, Qiu et al., 2025, Aghakhani et al., 2025, Wang et al., 2026, Sungarda et al., 2025, Liu et al., 2023, Chiu et al., 2024, Li et al., 2024, Wang et al., 2024a, Fouda et al., 2026, Cai et al., 2026, Qiu et al., 2026, Nguyen et al., 2025, Zhang et al., 2025a, Kim et al., 2025a, Xu et al., 2025a, Ozgun et al., 2025, Louie et al., 2026, Louie et al., 2024, Wang et al., 2024b, Lee et al., 2024, Basar et al., 2024, Yang et al., 2025b, Yang et al., 2025a, Basar et al., 2025, Feng et al., 2025, Kim et al., 2025c, Kim et al., 2025b, Samuel et al., 2025, Chen et al., 2025, Na, 2024, Sharma et al., 2023, Kang et al., 2024, Wang et al., 2025a
Custom expert rating <i>Expert judgments on custom dimensions</i>	Xiao et al., 2024, Gabriel et al., 2024, Haydarov et al., 2025, Wang et al., 2026, Qiu et al., 2026, Kim et al., 2025a, Louie et al., 2026, Louie et al., 2024, Wang et al., 2024b, Basar et al., 2024, Yang et al., 2025b, Basar et al., 2025, Kim et al., 2025c, Liu et al., 2025, Na, 2024, Sharma et al., 2023, Kang et al., 2024, Schmidgall et al., 2025, Wang et al., 2025a
Affective/emotional dynamics metric <i>Emotion recognition or affective computing metrics</i>	Haydarov et al., 2025, Schmidmaier et al., 2025a, Wang et al., 2025d
Clinical diagnostic benchmark <i>Clinical diagnosis or screening benchmarks</i>	Qiu et al., 2025, Hu et al., 2025, Fouda et al., 2026, Cai et al., 2026, Ozgun et al., 2025, Liu et al., 2025, Chen et al., 2023, Schmidgall et al., 2025
Theory-specific benchmark/task <i>Benchmark or task whose scoring criteria or target behaviors are derived from a psychological, clinical, or counseling construct.</i>	Qiu et al., 2025, Aghakhani et al., 2025, Wang et al., 2026, Sungarda et al., 2025, Chiu et al., 2024, Hu et al., 2025, Fouda et al., 2026, Nguyen et al., 2025, Zhang et al., 2025a, Kim et al., 2025a, Jaiswal et al., 2024, Xu et al., 2025a, Ozgun et al., 2025, Wang et al., 2024b, Lee et al., 2024, Kermani et al., 2025, Feng et al., 2025, Kim et al., 2025b, Liu et al., 2025, Wang et al., 2025d, Samuel et al., 2025, Na, 2024, Sharma et al., 2023, Kang et al., 2024, Chen et al., 2023, Tanana et al., 2019

Table 21: Theory-related evaluation types identified across the reviewed studies. Papers may appear in multiple rows because categories are not mutually exclusive.